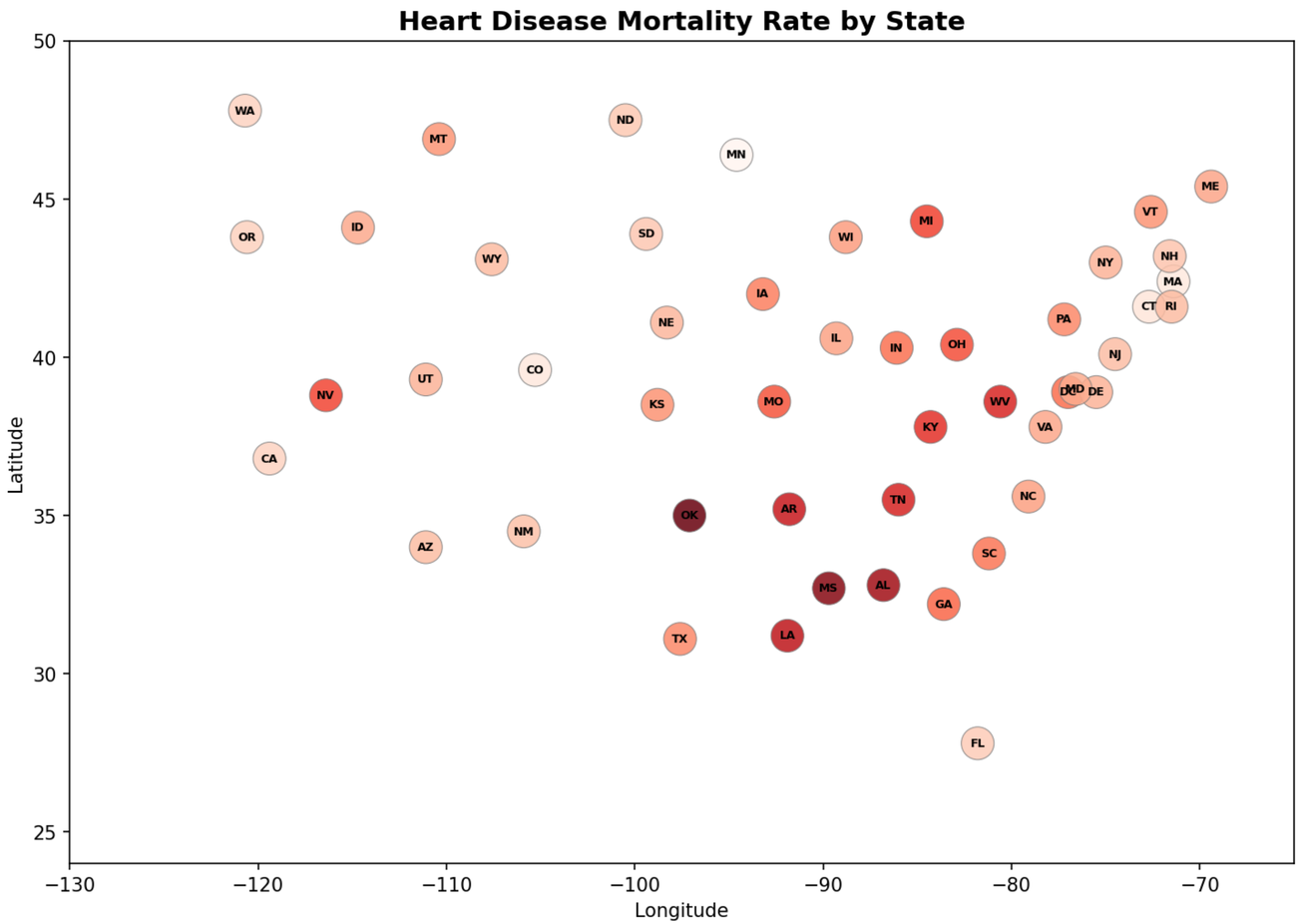
What Drives Chronic Disease Across U.S. States?

Predicting Heart Disease, Diabetes, Cancer & Mental Health from CDC Data

Jonathan Chamberlin | Min Yu Huang | Anuhya Mandava | Tsion Teklaeb

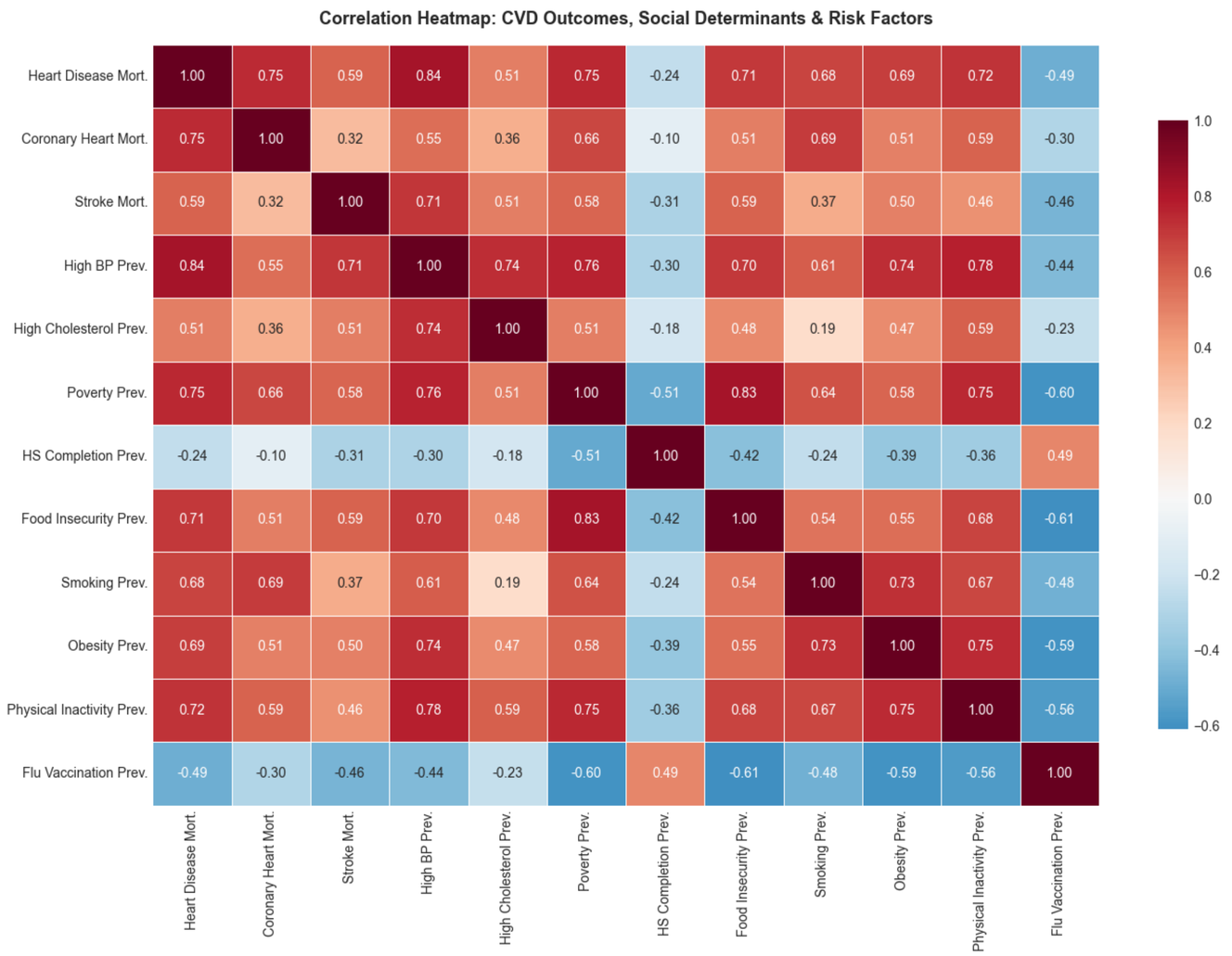
Can We Predict Where Chronic Disease Hits Hardest?

- Which factors best predict state-level disease rates?
- CDC Chronic Disease Indicators: 50 states + DC, 27 health metrics
- Federal government's official chronic disease tracking system
- Goal: target interventions where they'll have the most impact



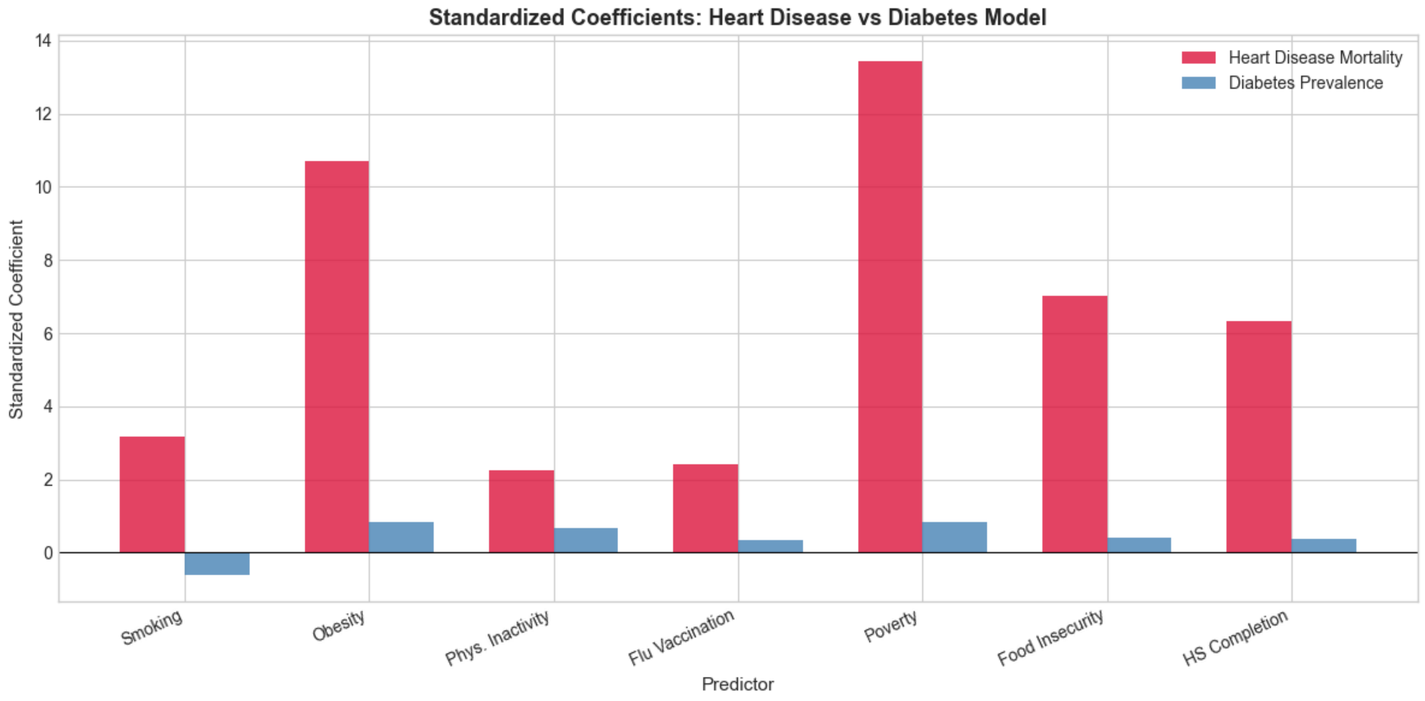
How We Analyzed the Data

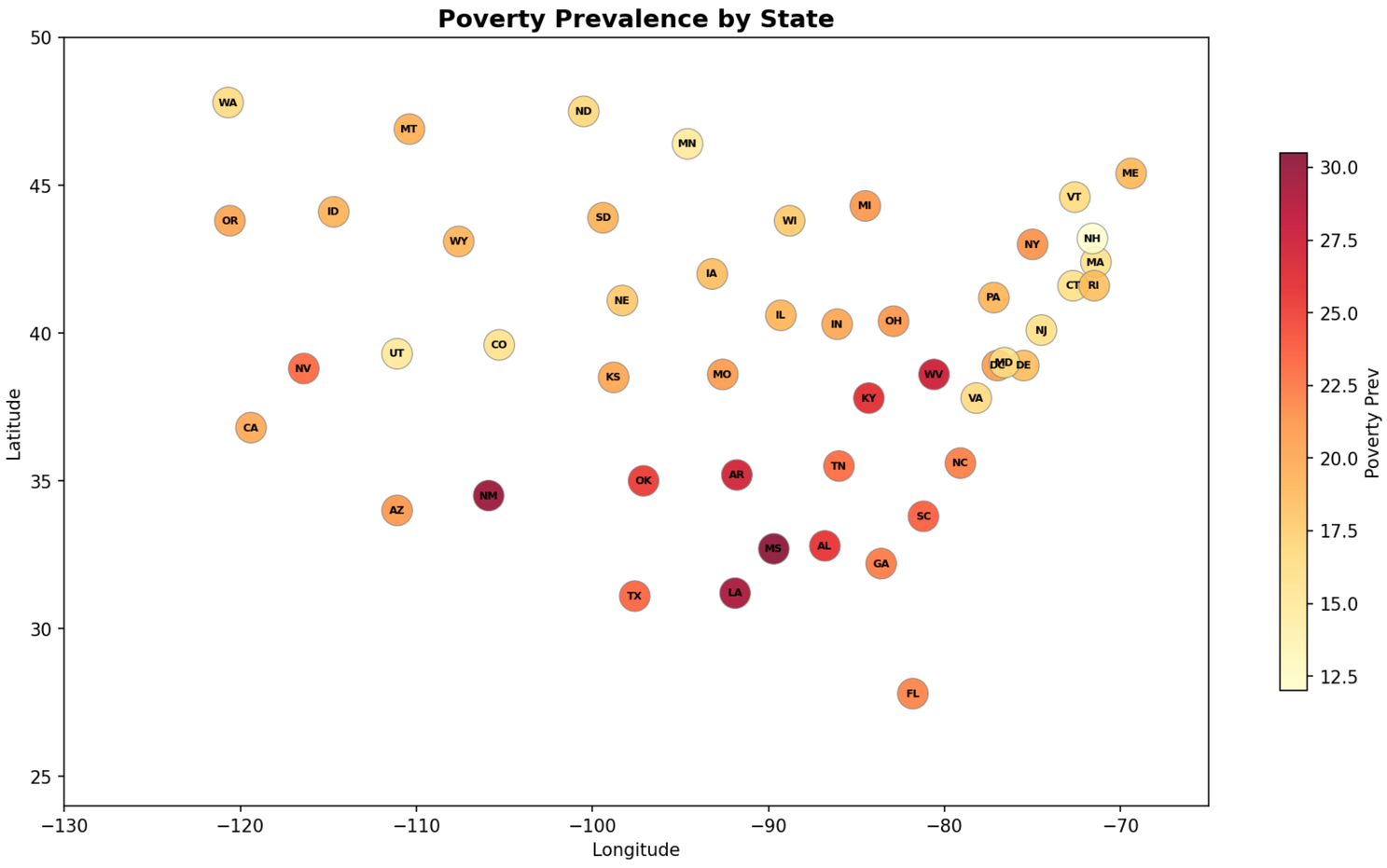
- Each team member analyzed a different disease pair
- Correlation analysis: which factors move together across states?
- Two prediction models: direct relationships + complex patterns
- Validated by predicting each state using only the other 50



Poverty Predicts Heart Disease More Than Smoking

- Best model explains 65% of state-level variation in heart disease mortality
- Poverty is the #1 predictor — nearly 4x larger effect than smoking
- States above median poverty: significantly higher CVD mortality ($p < 0.0001$)
- Diabetes tells a different story: obesity, not poverty, is the top driver



Conclusions

- Economic factors predict chronic disease as strongly as lifestyle factors
- Southeast clustering of heart disease + poverty: they reinforce each other
- Public health strategies targeting only individual behavior miss the larger picture
- Limitation: state-level data (n=51) cannot tell us about individuals